Parallelogram and its properties

One definition — both pairs of opposite sides parallel — and three properties that follow from a single pair of congruent triangles.

LESSON 5 | 1 × 40 MIN | GEOMETRY 8, TEMA 2 | STAGE 9 · BEYOND

LESSON CLOCK

Warm-up	4 min
Explanation	13 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- State the definition of a parallelogram.
- Prove and use: opposite sides equal, opposite angles equal, diagonals bisect each other.
- Use the fact that neighbouring angles are supplementary.
- Solve numerical problems on sides, angles and perimeter.

TEXTBOOK

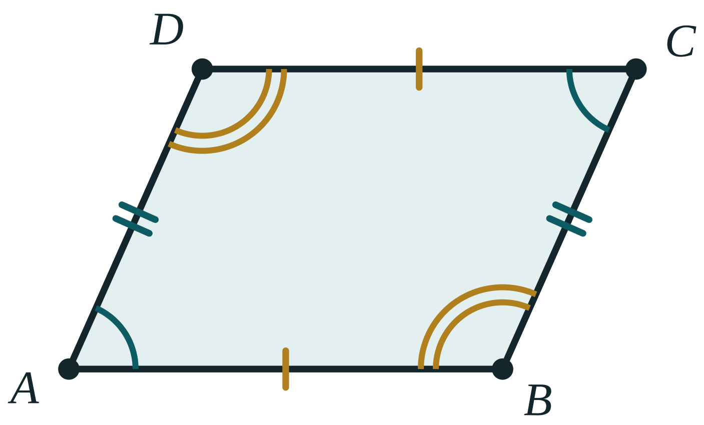
National	Тема 2, pp. 8–10
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

The definition

DEFINITION

A **parallelogram** is a quadrilateral whose opposite sides are pairwise parallel: $AB \parallel DC$ and $AD \parallel BC$.



Opposite sides carry the same tick marks; opposite angles carry the same arcs. Neighbouring angles add to 180°.

Notice what the definition does *not* say. It says nothing about lengths and nothing about angles. Everything else has to be proved.

Property 1 — opposite sides are equal

Proof. Draw the diagonal AC. It cuts the parallelogram into $\triangle ABC$ and $\triangle CDA$.

- AC is common to both.
- $\angle BAC = \angle DCA$ — alternate angles, since $AB \parallel DC$.
- $\angle BCA = \angle DAC$ — alternate angles, since $AD \parallel BC$.

So $\triangle ABC \cong \triangle CDA$ by ASA, and therefore $AB = CD$ and $BC = AD$. ∎

CONSEQUENCE

Perimeter $P = 2(a + b)$, where a and b are two neighbouring sides.

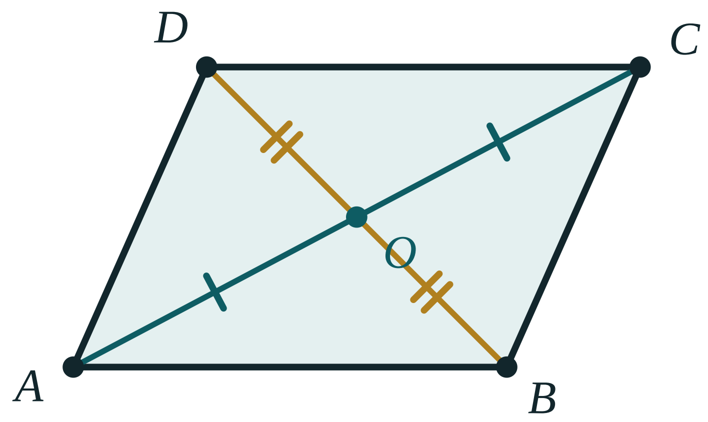
Property 2 — opposite angles are equal

The same congruent triangles give $\angle B = \angle D$ directly. For $\angle A$ and $\angle C$, add the two pairs of alternate angles at each of them.

AND NEIGHBOURING ANGLES

$\angle A + \angle B = 180°$ — they are co-interior angles between the parallels AD and BC. So knowing **one** angle of a parallelogram gives you all four.

Property 3 — the diagonals bisect each other



The diagonals cross at O, and each is cut into two equal parts: AO = OC and BO = OD.

Proof. Let the diagonals meet at O. In $\triangle AOB$ and $\triangle COD$:

- $AB = CD$ — Property 1.
- $\angle OAB = \angle OCD$ and $\angle OBA = \angle ODC$ — alternate angles on $AB \parallel DC$.

So $\triangle AOB \cong \triangle COD$ by ASA, giving $AO = OC$ and $BO = OD$. ∎

> BISECT, NOT EQUAL
>
> The diagonals cut **each other** in half. They are **not** equal to one another — that only happens in a rectangle, and it is the single most common error in this chapter.

02 Worked examples

EXAMPLE 1 ***In parallelogram ABCD, $\angle A = 65°$. Find the other three angles.***

1. $\angle C = \angle A = 65°$

   Opposite angles are equal.

2. $\angle B = 180° - 65° = 115°$

   Neighbouring angles are supplementary.

3. $\angle D = \angle B = 115°$

   Opposite angles again.

4. *check:* $65 + 115 + 65 + 115 = 360$ ✓

   The angle sum of a quadrilateral.

ANSWER

$\angle B = \angle D = 115°$, $\angle C = 65°$

EXAMPLE 2

The perimeter of a parallelogram is 48 cm and one side is 3 cm longer than the other. Find the sides.

1. $2(a + b) = 48$, so $a + b = 24$

   Perimeter formula.

2. $b = a + 3$

   The given relation.

3. $a + a + 3 = 24 \implies 2a = 21 \implies a = 10.5$

4. $b = 13.5$

ANSWER

$10.5\ cm$ and $13.5\ cm$

EXAMPLE 3

The diagonals of parallelogram ABCD meet at O, with AC = 14 cm and BD = 10 cm. Find AO and OD.

1. $AO = OC = \frac{14}{2} = 7\ cm$

   The diagonals bisect each other.

2. $BO = OD = \frac{10}{2} = 5\ cm$

   Same property, other diagonal.

3. Note that $AC \neq BD$ — the diagonals are not equal here.

   They would be only in a rectangle.

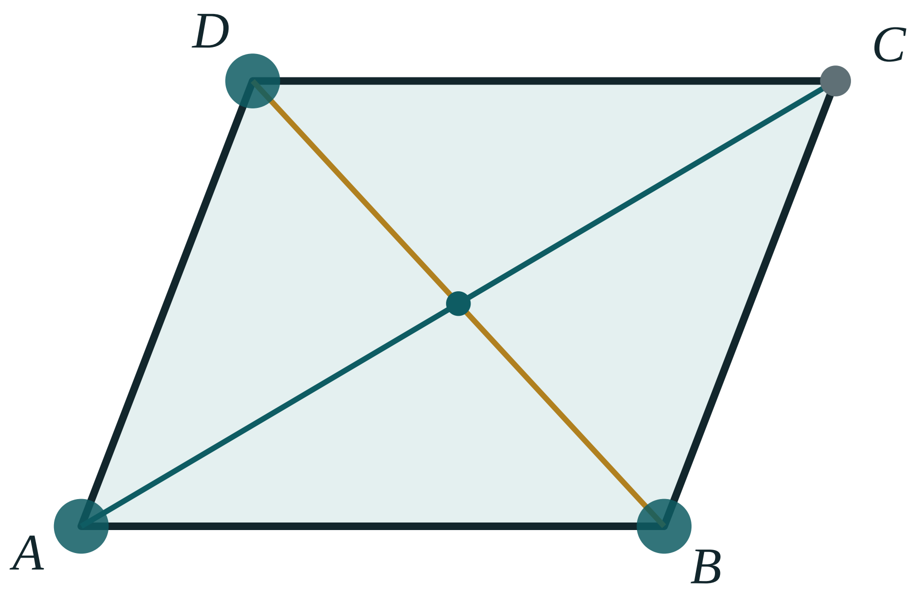
ANSWER

$AO = 7\ cm,\ OD = 5\ cm$

03 Interactive model

Drag A, B or D and watch AB = DC and AD = BC hold, while AC and BD stay different.

Drag the parallelogram



AB **170** | DC **170** | AD **139.3** | BC **139.3** | ∠A **69°** | ∠C **69°** | AC **255.5**

BD **176.9** | AO **127.8** | OC **127.8**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Parallelogram	Parallelogramm	Параллелограмм
Opposite sides	Qarama-qarshi tomonlar	Противоположные стороны
Opposite angles	Qarama-qarshi burchaklar	Противоположные углы
Adjacent angles	Qo‘shni burchaklar	Прилежащие углы
The diagonals bisect each other	Diagonallar bir-birini teng ikkiga bo‘ladi	Диагонали делятся пополам
Perimeter	Perimetr	Периметр
Property	Xossa	Свойство
Proof	Isbot	Доказательство

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

In a parallelogram, the diagonals:

- are equal
- bisect each other
- are perpendicular
- bisect the angles

$\angle A = 110°$ in parallelogram ABCD. Then $\angle B$ is:

- $110°$
- $70°$
- $55°$
- $90°$

A parallelogram has sides 7 cm and 5 cm. Its perimeter is:

- 12 cm
- 24 cm
- 35 cm
- 17 cm

The proof that opposite sides are equal uses:

- SSS
- ASA with a diagonal
- RHS
- the angle sum of a quadrilateral

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *In parallelogram ABCD, $AB = 8$ cm. Find CD.*

   ANSWER *8 cm*

2. $\angle A = 70°$. *Find* $\angle C$.

   ANSWER *70°*

3. $\angle A = 70°$. *Find* $\angle B$.

   ANSWER *110°*

4. *Sides 6 cm and 9 cm. Find the perimeter.*

   ANSWER *30 cm*

5. $AC = 12$ *cm and the diagonals meet at O. Find AO.*

   ANSWER *6 cm*

6. $BD = 9$ *cm. Find OD.*

   ANSWER *4.5 cm*

7. *One angle of a parallelogram is 90°. What are the others?*

   ANSWER *90°* each — it is a rectangle.

Medium · the standard question

7 PROBLEMS

1. $\angle A = 65°$. *Find all four angles.*

   ANSWER $65°, 115°, 65°, 115°$

2. *The perimeter is 48 cm and one side is 3 cm longer than the other. Find the sides.*

   ANSWER *10.5 cm* and *13.5 cm*

3. $AC = 14$ *cm*, $BD = 10$ *cm. Find* AO *and* OD.

   ANSWER $AO = 7$ *cm*, $OD = 5$ *cm*

4. *Two neighbouring angles are in the ratio* $2 : 3$. *Find all four angles.*

   ANSWER $72°, 108°, 72°, 108°$

5. *The perimeter is 36 cm and one side is twice the other. Find the sides.*

   ANSWER *6 cm* and *12 cm*

6. $\angle A - \angle B = 40°$. *Find both angles.*

   ANSWER $\angle A = 110°, \angle B = 70°$

7. *In parallelogram* $ABCD$, $AO = 5$ *cm and* $OD = 3$ *cm. Find* AC *and* BD.

   ANSWER $AC = 10$ *cm*, $BD = 6$ *cm*

Hard · stretch and reason

7 PROBLEMS

1. *The bisector of $\angle A$ of parallelogram ABCD meets BC at K. Prove that $BK = AB$.*

   ANSWER $\angle BAK = \angle KAD = \angle AKB$ (alternate angles), so $\triangle ABK$ is isosceles and $BK = AB$.

2. *In parallelogram ABCD, $AB = 6$, $BC = 10$, and the bisector of $\angle A$ meets BC at K. Find KC.*

   ANSWER $BK = AB = 6$, so $KC = 10 - 6 = 4$.

3. *Prove that a parallelogram with one right angle is a rectangle.*

   ANSWER Neighbouring angles are supplementary, so the neighbours are $90°$ too, and opposite angles equal them.

4. *The diagonals of a parallelogram are 16 cm and 12 cm and one side is 10 cm. Find the perimeter, given the diagonals are perpendicular.*

   ANSWER Perpendicular diagonals make it a rhombus, so all sides are $10\ cm$ and $P = 40\ cm$.

5. *In parallelogram ABCD the perpendicular from B to AD has length 6 and $AD = 9$. Find the area.*

   ANSWER $S = 9 \cdot 6 = 54$ square units.

6. *Prove that if the diagonals of a quadrilateral bisect each other, the opposite sides are equal.*

   ANSWER $\triangle AOB \cong \triangle COD$ by SAS (vertically opposite angles), so $AB = CD$; similarly $BC = AD$.

7. *In parallelogram ABCD, $\angle A$ is three times $\angle B$. Find all four angles.*

   ANSWER $\angle A = \angle C = 135°$, $\angle B = \angle D = 45°$

07 Homework

Homework — 5 problems

Geometry 8, Тема 2, pp. 8–10. Draw and label a figure for each.

1. *In parallelogram ABCD, $\angle D = 118°$. Find the other three angles.*
2. *The perimeter is 54 cm and one side is 5 cm shorter than the other. Find the sides.*
3. *The diagonals meet at O with $AO = 6.5$ cm and $BO = 4$ cm. Find AC and BD.*

4. *Two neighbouring angles are in the ratio 4 : 5. Find all four.*

5. *Prove that the diagonal of a parallelogram divides it into two congruent triangles.*